On the Round Complexity of Dishonest-Majority MPC

Ran Cohen, Daniel Collins, **Pouyan Forghani**, Juan Garay, and Vassilis Zikas

Reichman University, NYU & Hebrew University, Georgia Tech, Texas A&M University

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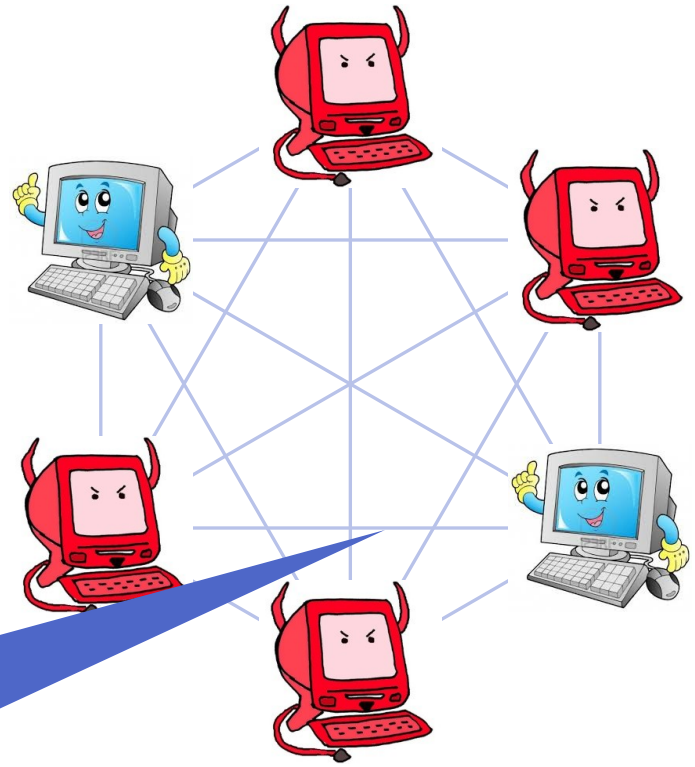
Secure Multiparty Computation (MPC)

This work:

Dishonest majority $t \geq n/2$

- Fairness is impossible in general
- Security *with abort*

Main question:
What is the round complexity of MPC?



Communication over **point-to-point** channels only!

Three Flavors of Abort

Selective (SA)



y $\perp$ y

Some may get y
others may get $\perp$

<

Unanimous (UA)



$\perp$ $\perp$ $\perp$

Either all get y
or all abort

<

Identifiable (IA)



$(\perp, P_{i^*})$ $(\perp, P_{i^*})$ $(\perp, P_{i^*})$

Either all get y
or all abort & identify P_{i^*}

So, what's the round complexity of MPC over P2P?

Recipe for MPC over Point-to-Point ($t < n/2$)



MPC over broadcast ✦ Broadcast protocol → MPC over P2P

Broadcast protocol [PSL80, LSP82]
Sender P_s holds input x
Agreement: honest agree on output
Validity: honest sender $\Longrightarrow$ output x

Honest majority [CCGZ16]:
- Same expected rounds
- Same security (G.O.D)

Recipe for MPC over Point-to-Point ($t \geq n/2$)



UA-MPC over broadcast ✦ Broadcast protocol ➡ UA-MPC over P2P

Security is preserved
What about **rounds**?

Round complexity of broadcast $t \geq n/2$

Deterministic:

- $t + 1$ rounds sufficient & necessary [DS83]

Randomized:

- Expected $O(1)$ rounds for $t = (1 - \epsilon)n$ [WXSD20]
- $\Omega(n/(n - t))$ rounds [GKKO07]

Let's Plug & Play



UA-MPC [CGZ20]
- 1st round P2P
- 2nd round BC

✦

Broadcast [WXSD20]
- Exp $O(1)$

→

UA-MPC over P2P
- Exp $O(\log n)$

Idea: consider broadcast with abort

Everyone broadcast in 2nd round
Requires parallel composition



Where do we stand?

Upper bounds

- Exp $O(1)$ round BC **only** for $t = (1 - \epsilon)n$
- No round-preserving parallel composition

Lower bounds

- LB for broadcast assume **guaranteed output delivery** [DS83, CMS89, KY84, GKKO07]
- Do not apply for broadcast with abort

So, what's the round complexity of MPC over P2P with **UA** or **IA**?

This Work

MPC	Strict $O(1)$ rounds $t=\omega(1)$	Expected $O(1)$ rounds $t=(1-\varepsilon)n$	Expected $O(1)$ rounds $t=n-o(n)$
SA	✓ 2 rounds [CGZ20]	✓	✓
UA	✗ **lower bound**	✓ **OT ✦ VRFs, PKI**	?
IA	✗ **follows from UA**	?	✗ **lower bound**

	Strict	Exp. $(1-\varepsilon)n$	Exp. $n-o(n)$
UA	✗		
IA	✗		

UA Lower Bound

Thm: No constant-round UA-broadcast for $t = \omega(1)$

Follows directly from the hybrid chain of scenarios [CMS89, KY84]

S_1 all honest, input 0 $\equiv$ S_2 $\leq t$ corrupted $\equiv \cdots \equiv$ S_{k-1} $\leq t$ corrupted $\equiv$ S_k all honest, input 1

No abort, output 0 ➡ No abort, output 0 ... ➡ No abort, output 0 ➡ No abort, output 0

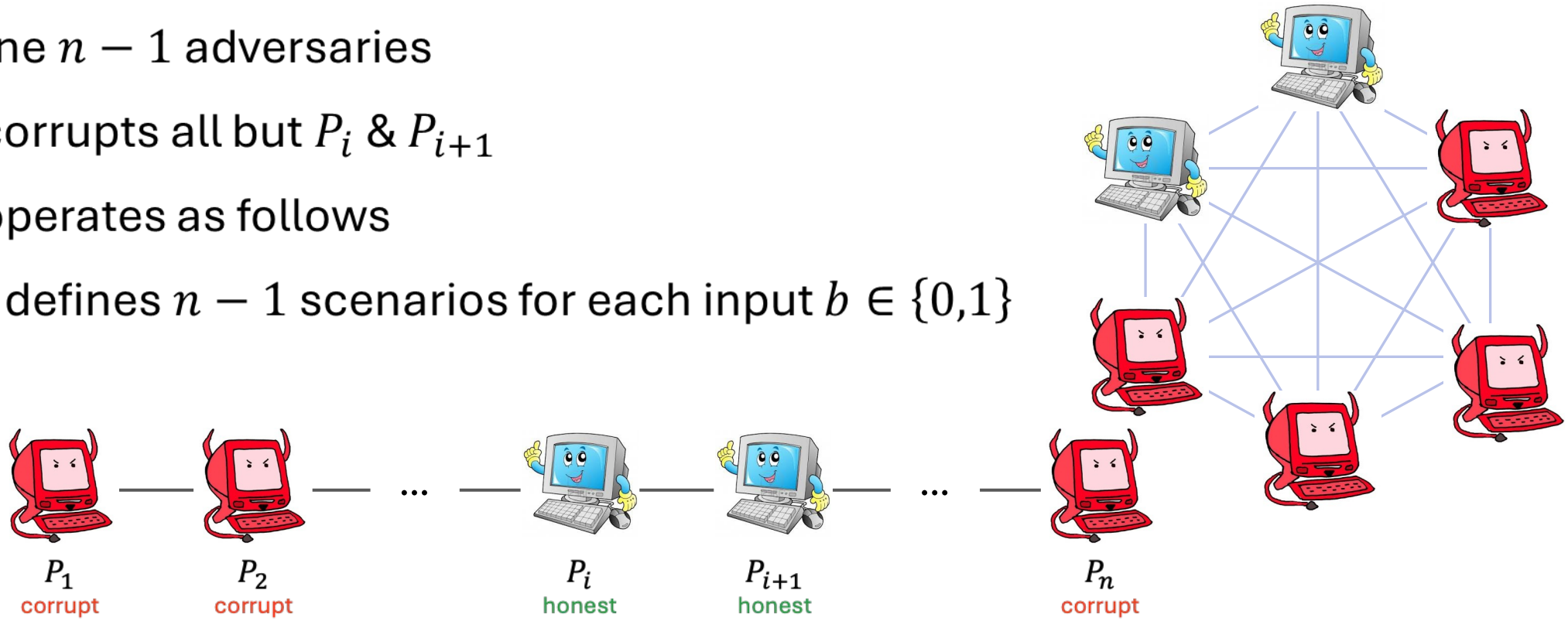
No abort, output 1

- No abort in all honest executions
- Same view distributions $\Longrightarrow$ output 0 in all scenario
- Classical bound: error $\varepsilon \geq \frac{1}{2}\left(\frac{t}{2nr}\right)^r$, non-negligible for $r = O(1)$

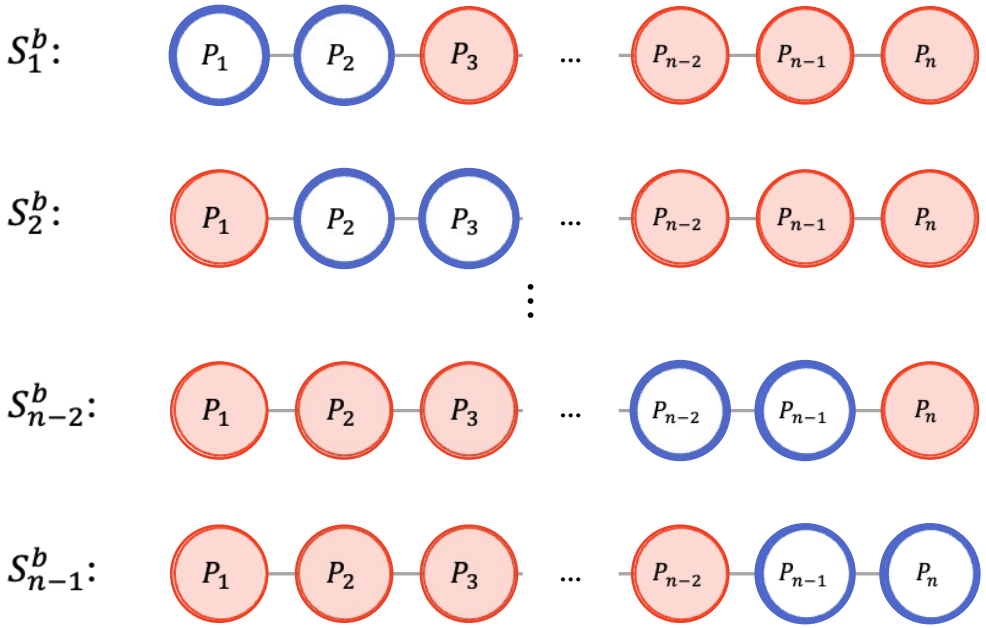
Recall [GKKO01] Lower Bound for Broadcast

[GKKO07]: broadcast with guaranteed output needs $\Omega(n/(n-t))$ rounds

- Define $n-1$ adversaries
- $\mathcal{A}_i$ corrupts all but P_i & P_{i+1}
- $\mathcal{A}_i$ operates as follows
- This defines $n-1$ scenarios for each input $b \in \{0,1\}$



Recall [GKKO01] Lower Bound for Broadcast



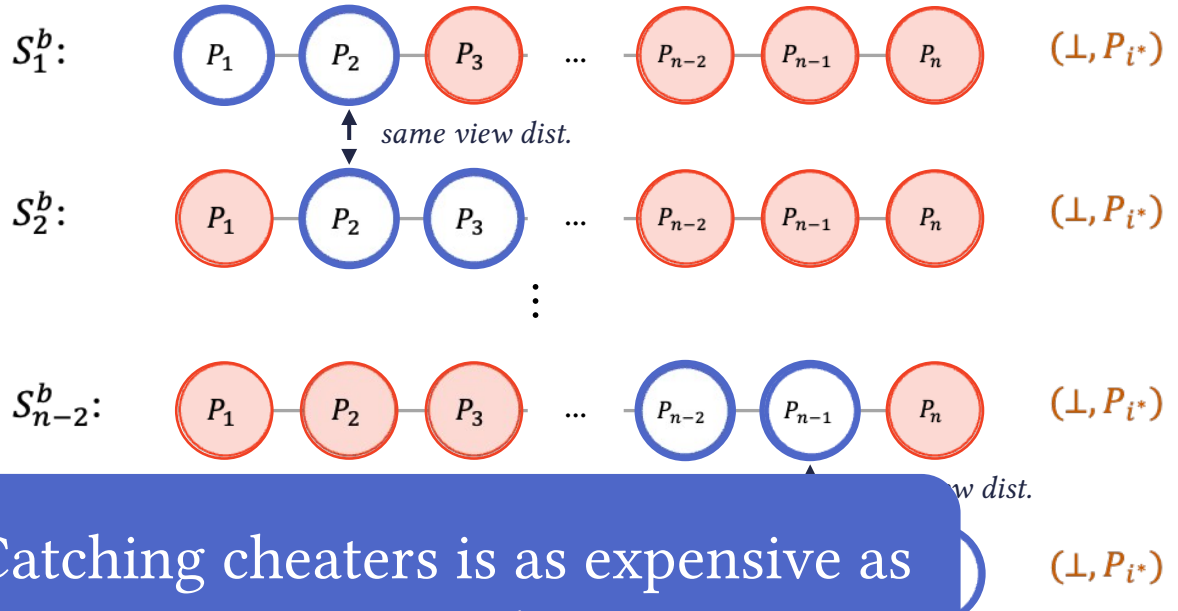
- Neighbors share an honest view
  $\Longrightarrow$ all agree along the path
- One hop per round
  $\Longrightarrow P_n$ needs $n-1$ rounds

Argument doesn't apply to UA
Everyone immediately detects an attack

What about IA?

	Strict	Exp. $(1-\varepsilon)n$	Exp. $n-o(n)$
UA	✗		
IA	✗		✗

Catching Cheaters Costs Rounds



Catching cheaters is as expensive as overcoming cheating!

- Suppose honest identify P_{i^*} in S_{n-1}^b
- Same view distribution
  $\Longrightarrow P_{i^*}$ identified in all scenarios
- But every party is honest in some scenario $\Longrightarrow$ contradiction
- No abort
  $\Longrightarrow$ the [GKKO07] argument applies

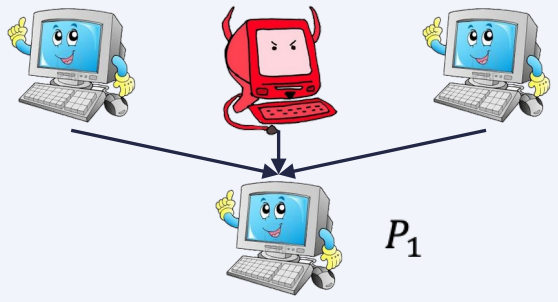
Thm: Every t-secure IA-broadcast needs $\geq \frac{2n}{n-t} - 1$ rounds
$\Longrightarrow$ no expected-constant-round IA-MPC for $t = n - o(n)$

	Strict	Exp. $(1-\varepsilon)n$	Exp. $n-o(n)$
UA	✗	?	
IA	✗		✗

Upper Bound: One Broadcast per Round

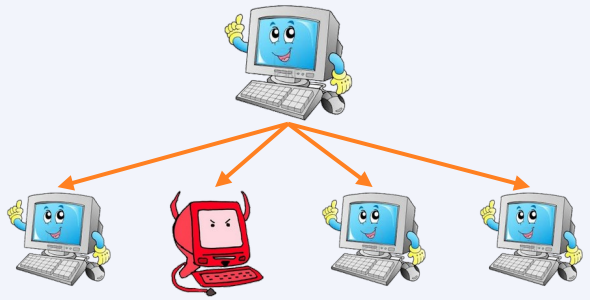
Our compiler: single-sender $\Longrightarrow$ parallel UA-broadcast, black-box

Sign, send to P_1



$\sigma_i = \mathrm{Sign}_{sk_i}(s, i, x_i)$

P_1 broadcasts *once*



$((x_1, \sigma_1), \ldots, (x_n, \sigma_n))$

Verify all signatures

All valid $\Longrightarrow$
output $(x_1, \ldots, x_n)$

Any invalid $\Longrightarrow$
abort

- Same broadcast output
  $\Longrightarrow$ honest parties abort together
- Signatures
  $\Longrightarrow$ a corrupt P_1 can't alter honest inputs

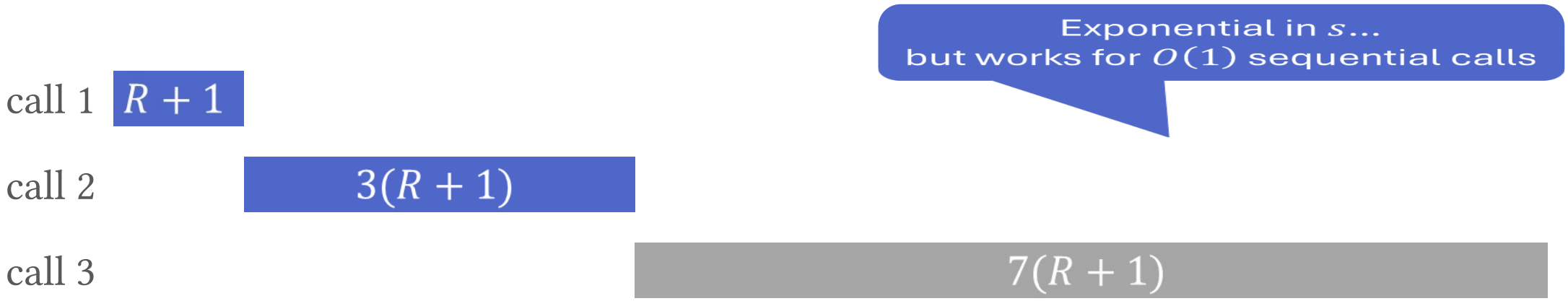
One BC instead of n:
no max-of-n blowup!

Not IA: was it P_j or P_1?

	Strict	Exp. $(1-\varepsilon)n$	Exp. $n-o(n)$
UA	✗	?	
IA	✗		✗

Careful: Sequential Calls Drift Apart

- Broadcast doesn't end simultaneously ⟹ next call starts out of sync
- Honest-majority techniques a la [Bracha84] do not apply
- Round expansion [LLR02]: in call s, stretch each round to $2^s - 1$ rounds
- Call s takes expected $(2^s - 1)(R + 1)$ rounds



	Strict	Exp. $(1-\varepsilon)n$	Exp. $n-o(n)$
UA	✗	✓	
IA	✗		✗

Expected Constant Rounds? Yes!

2-round UA-MPC over Broadcast [GS18, BL18] ✦ **Our compiler** ✦ UA-broadcast in exp $O(1)$ rounds [WXSD25]

OT signatures, PKI VRFs, PKI

$$\underbrace{(R+1)}_{\text{call 1}} + \underbrace{3(R+1)}_{\text{call 2}} = \underbrace{4(R+1) = O(1)}_{\text{expected rounds over P2P}}$$

Thm: OT and VRFs $\Longrightarrow$
UA-MPC in PKI model over P2P in exp $O(1)$ rounds, for $t = (1-\varepsilon)n$

Summary

MPC	Strict $O(1)$ rounds $t=\omega(1)$	Expected $O(1)$ rounds $t=(1-\varepsilon)n$	Expected $O(1)$ rounds $t=n-o(n)$
SA	✓ 2 rounds [CGZ20]	✓	✓
UA	✗ **lower bound**	✓ **OT ✦ VRFs, PKI**	✗ follow-up [CFGZ26]
IA	✗ **follows from UA**	✓ concurrent [CEG26]	✗ **lower bound**

UA: one broadcast per round suffices $\Longrightarrow$ expected $O(1)$ rounds

IA: catching cheaters costs as many rounds as overcoming cheating

Thank you!

https://eprint.iacr.org/2026/1277